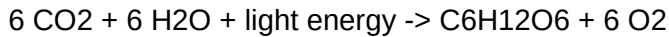


BIOL 110 - Lecture 12: Photosynthesis

Learning objectives

- Describe the light-dependent reactions and the Calvin cycle.
- Explain how C4 and CAM plants reduce photorespiration.

Overview



Photosynthesis happens in chloroplasts. The thylakoid membranes hold the pigments; the fluid around them is the stroma.

Light-dependent reactions (thylakoid membrane)

- Photosystem II absorbs light and splits water, releasing oxygen as a by-product.
- Electrons flow through the electron transport chain to Photosystem I.
- The proton gradient drives ATP synthase to make ATP.
- Photosystem I passes electrons to NADP⁺, forming NADPH.

Calvin cycle (stroma)

- The enzyme RuBisCO fixes CO₂ onto ribulose biphosphate (RuBP).
- ATP and NADPH from the light reactions reduce the products to G3P sugar.
- Most G3P regenerates RuBP; one in six molecules leaves to build glucose.

Photorespiration

RuBisCO sometimes grabs O₂ instead of CO₂, wasting energy, especially when hot and dry.

C4 plants (corn, sugarcane) concentrate CO₂ in bundle sheath cells.

CAM plants (cacti, pineapple) open stomata only at night to save water.

Exam reminder: know the inputs and outputs of each stage.